

PALERMO PUNTA RAISI, Italy

WMO: 164050

Lat: 38.182N

Lon: 13.099E

Elev: 34

StdP: 100.92

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	7.8	-1.9	3.2	10.5	0.1	3.8	10.9	15.3	13.3	14.2	13.3	4.4	170	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	33.1	22.3	31.1	23.0	29.9	23.6	26.3	28.9	25.7	28.6	25.0	28.1	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.2	28.2	24.8	19.9	27.7	24.0	18.9	27.4	82.4	28.9	79.0	28.7	76.1	28.2	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.6	10.3	DB	4.3	38.0	1.7	2.1	3.1	39.5	2.1	40.8	1.1	42.0	-0.1	43.5	(4)
(5)			WB	2.3	27.5	1.4	1.7	1.3	28.7	0.5	29.7	-0.3	30.6	-1.3	31.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	12.4	12.2	13.7	15.9	19.2	23.2	26.0	26.6	23.9	20.8	17.2	14.0	(6)
(7)		DBStd	5.55	2.28	2.41	2.43	2.38	2.51	2.54	1.75	1.84	2.19	2.24	2.61	2.35	(7)
(8)		HDD10.0	13	4	6	2	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	792	183	173	147	80	16	0	0	0	4	51	137		(9)
(10)		CDD10.0	3226	81	67	116	178	287	396	495	515	418	334	216	124	(10)
(11)		CDD18.3	962	1	1	3	8	44	146	237	257	168	79	17	2	(11)
(12)		CDH23.3	7091	0	3	21	39	167	819	2116	2556	1069	250	49	2	(12)
(13)		CDH26.7	1797	0	0	5	5	49	201	531	743	204	53	5	0	(13)
(14)	Wind	WSAvg	4.4	5.2	5.3	4.9	4.4	4.0	3.9	3.7	3.6	4.1	4.0	4.9	5.3	(14)
(15)	Precipitation	PrecAvg	622	108	80	74	66	29	12	5	18	81	110	113	99	(15)
(16)		PrecMax	1014	458	386	401	163	113	37	17	90	375	432	330	168	(16)
(17)		PrecMin	396	16	8	0	8	3	0	0	1	8	0	28	65	(17)
(18)		PrecStd	138	110	82	86	37	28	11	5	23	88	111	78	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	22.9	26.0	27.2	33.0	35.9	35.0	36.2	33.8	31.8	27.2	23.1	(19)
(20)			MCWB	12.6	13.5	14.7	15.5	18.0	19.1	22.3	22.7	21.4	18.6	16.9	14.3	(20)
(21)		2%	DB	18.3	18.9	21.8	24.0	27.0	31.0	32.0	33.0	30.6	27.9	24.7	20.2	(21)
(22)			MCWB	12.9	12.7	14.1	15.5	18.3	21.1	22.5	23.6	22.1	20.0	16.5	14.3	(22)
(23)		5%	DB	17.0	17.0	18.8	21.2	24.8	28.7	30.2	31.0	28.8	25.8	22.8	18.7	(23)
(24)			MCWB	13.0	12.7	13.3	14.9	18.2	21.5	23.6	24.3	22.7	20.2	16.5	14.0	(24)
(25)		10%	DB	15.9	15.8	17.1	19.6	23.0	27.1	29.1	29.8	27.3	24.2	21.1	17.2	(25)
(26)			MCWB	12.6	12.4	13.3	14.9	18.3	21.8	23.7	24.5	22.3	19.9	16.5	13.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.1	16.2	17.9	21.2	25.5	27.0	27.8	25.9	23.5	19.7	16.4	(27)
(28)			MCD B	17.5	18.4	20.4	22.0	24.0	27.9	29.1	29.8	28.5	25.5	22.8	19.0	(28)
(29)		2%	WB	14.1	14.1	15.1	16.9	20.4	24.2	26.0	26.8	24.7	22.0	18.5	15.5	(29)
(30)			MCD B	16.6	16.9	18.2	19.9	23.2	27.0	28.8	29.3	27.6	24.8	21.8	18.3	(30)
(31)		5%	WB	13.4	13.3	14.3	16.2	19.7	23.4	25.3	26.0	23.8	21.0	17.6	14.8	(31)
(32)			MCD B	16.0	15.9	17.1	19.0	22.5	26.3	28.3	28.9	27.1	24.2	20.9	17.7	(32)
(33)		10%	WB	12.7	12.6	13.7	15.6	18.9	22.6	24.7	25.4	22.9	20.4	16.9	14.1	(33)
(34)			MCD B	15.5	15.2	16.5	18.4	21.8	25.6	27.8	28.5	26.6	23.6	20.1	17.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.4	5.9	6.2	6.2	6.3	6.2	5.7	6.0	5.5	4.9	(35)	
(36)			MCD BR	5.8	6.8	8.3	9.2	9.3	9.3	8.3	8.3	7.4	7.6	7.2	5.9	(36)
(37)			MCWBR	4.2	4.5	4.7	5.0	4.8	5.3	5.6	5.8	5.3	4.6	4.6	4.0	(37)
(38)		5% WB	MCD BR	5.0	6.0	6.7	7.2	7.3	6.9	6.8	7.0	6.4	6.2	5.9	5.0	(38)
(39)			MCWBR	4.3	4.5	4.7	4.8	4.7	5.0	5.3	5.6	5.1	4.5	4.4	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.368	0.407	0.445	0.451	0.428	0.430	0.430	0.436	0.410	0.378	0.353	(40)	
(41)		taud	2.451	2.370	2.254	2.163	2.191	2.303	2.307	2.310	2.295	2.344	2.382	2.428	(41)	
(42)		Ebn at Noon	827	853	849	836	836	854	849	841	811	794	777	787	(42)	
(43)		Edh at Noon	84	103	126	147	145	130	129	125	120	104	89	81	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.18	3.01	4.34	5.62	6.75	7.54	7.64	6.69	5.00	3.65	2.50	1.95	(44)	
(45)		RadStd	0.16	0.29	0.35	0.33	0.34	0.23	0.17	0.30	0.23	0.24	0.15	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	-0.84	-1.08	N/A	N/A	N/A	N/A	(46)	
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	-0.56	-0.65	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page